1 Appendix: Convergence Theoretical Analysis

We extend the theoretical analysis from Sancus [?] to the scenario of sampling-based graph embedding training. Our goal is to prove the convergence of a Graph Neural Network (GNN) model trained on sampled subgraphs with stale embeddings. We maintain the same system setup and assumptions as in the original work but adapt the analysis to account for sampling.

1.1 Preliminaries

Let $G_{ori} = (V, E)$ be a graph with adjacency matrix A_{ori}. The sampling process selects a subset of edges $S \subset E$ such that the sampled subgraph $G = (V_S, S)$ is representative of the entire graph, the adjacency matrix of G is A. Considering the multi-layer neighbor sampling, sampled subgraph contain multiple bipartite graphs, each responding to a layer, and we denote each bipartite graph with $A^{(\ell)}$. Although the $A^{(\ell)}$ in each iteration are different due to random sampling, but the ℓ_∞ norm of adjacency matrix and the number of nodes are similar and bounded, so we don't differentiate $A^{(\ell)}$ of each iteration in the following proof.

Consider a GNN model with L layers, initial embedding matrix X, and labels Y. The goal is to learn node representations $H^{(L)}$ that minimize the loss function $\mathcal{L}$ with respect to Y. The propagation process of such GNNs can be formulated in matrix form as:

$$H^{(\ell)} = \sigma\left(\hat{A} H^{(\ell-1)} W^{(\ell-1)}\right)$$

where $H^{(\ell)}$ is the embedding matrix at layer ℓ, σ denotes the activation function such as ReLU, $W^{(\ell)} \in \mathbb{R}^{F \times F}$ is the weight matrix, and $\hat{A}$ is the symmetrically normalized adjacency matrix. Initially, $H^{(0)} = X$ where $X \in \mathbb{R}^{N \times F}$ is the node embedding matrix.

For convenience, the superscript notation is omitted when it is clear from context. And we use T and Z to represent intermediate results:

$$T^{(\ell)} = \hat{A}^{(\ell)} H^{(\ell-1)}$$

$$Z^{(\ell)} = T^{(\ell)} W^{(\ell-1)}$$

$$H^{(\ell)} = \sigma(Z^{(\ell)})$$

As well as the backpropagation processes:

$$\delta^{(\ell)} = \frac{\partial \mathcal{L}}{\partial Z^{(\ell)}} = \delta^{(\ell+1)} \hat{A}^{(\ell+1)} (W^{(\ell)})^\intercal \odot \sigma'(Z^{(\ell)})$$

$$\nabla_{W^{(\ell)}} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \delta^{(\ell+1)} \hat{A}^{(\ell+1)} (H^{(\ell)})^\intercal$$

where $\delta^{(\ell)}$ is the gradient of the loss with respect to the input of layer ℓ, and σ' is the derivative of the activation function.

Taking into account the access patterns brought by sampling-based training, our approach implements low staleness for frequently accessed vertices while assigning higher staleness to those that are accessed less frequently. This strategy aims to distribute the error resulting from staleness more evenly across the vertices. We denote the staleness-induced error as ε.

1.2 Bounded Approximation Error of Intermediate Results and Gradients

Assumption 1 (Assumption on smooth activation function, gradent and loss function)**.** *The activation function $\sigma(\cdot)$ and the gradient $\nabla \mathcal{L}$ are ρ-Lipschitz continuous. The loss function $\mathcal{L}(W)$ is ρ-smooth.*

Remark. This assumption is commonly used and necessary within the framework of machine learning optimization theory [?] and holds for most situations. Most activations are smooth and loss function like MSE is also smooth.

Assumption 2 (Assumption on bounded matrices)**.** *The matrices $\hat{A}$, H, and W are bounded by some constant B.*

Remark. This holds for almost all training scenario because properly trained models don't contain infinite values, they are always bounded by some constant B.

Lemma 3. *For any input activations $\tilde{H}_i^{(l)}$ in each process $P(i)$ where $i \in \{1, ..., p\}$. With Assumption 1 and 2, Given the staleness bound ε_H where $\|\tilde{H}_i^{(l)} - H_i^{(l)}\| \leq \varepsilon_H$, the approximation error of the intermediate results $\tilde{T}_i^{(l)}$ and $\tilde{Z}_i^{(l)}$ is bounded by:*

$$\|\tilde{T}_i^{(l)} - T_i^{(l)}\|_\infty \leq pCB\varepsilon_H$$

$$\|\tilde{Z}_i^{(l)} - Z_i^{(l)}\|_\infty \leq C^2 B^2 \varepsilon_H$$

Here, C represents the maximum number of neighbors for vertices in $\hat{A}$. The matrices are obtained through sampling, and C is determined by the sampling configurations. In our evaluations, C is consistently smaller than 25.

Proof. By the properties of matrix multiplication and the triangle inequality, we have:

$$\|\tilde{T}_i^{(l)} - T_i^{(l)}\|_\infty = \|\sum_{j=1}^{p} \hat{A}_{ij}^{(\ell)} \tilde{H}_j^{(l-1)} - \sum_{j=1}^{p} \hat{A}_{ij}^{(\ell)} H_j^{(l-1)}\|_\infty$$

$$\leq \sum_{j=1}^{p} \|\hat{A}_{ij}^{(\ell)}\|_\infty \|\tilde{H}_j^{(l-1)} - H_j^{(l-1)}\|_\infty \leq pCB\varepsilon_H$$

Similarly, for $\tilde{Z}_i^{(l)}$:

$$\|\tilde{Z}_i^{(l)} - Z_i^{(l)}\|_\infty = \|\hat{A}_i^{(\ell)} \tilde{H}_i^{(l-1)} W_i^{(l-1)} - \hat{A}_i^{(\ell)} H_i^{(l-1)} W_i^{(l-1)}\|_\infty$$

$$\leq \|\hat{A}_i^{(\ell)}\|_\infty \|\tilde{H}_i^{(l-1)} - H_i^{(l-1)}\|_\infty \|W_i^{(l-1)}\|_\infty \leq C^2 B^2 \varepsilon_H$$

□

Lemma 4. *For each process $P(i)$ where $i \in [1,p]$, with Assumption 1 and 2, the approximation error of the gradients $\nabla_{\tilde{Z}_i^{(\ell)}} \tilde{\mathcal{L}}$ and $\nabla_{W_i^{(\ell)}} \tilde{\mathcal{L}}$ is bounded by:*

$$\|\nabla_{\tilde{Z}_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(\ell)}} \mathcal{L}\|_\infty \leq K$$

$$\|\nabla_{W_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{W_i^{(\ell)}} \mathcal{L}\|_\infty \leq K$$

where K depends on ρ, C, B, and ε_H.

Proof. We will use induction to prove the bounds on the approximation errors of the gradients.

Base Case: For the final layer L, By the Lipschitz continuity of $\sigma'(\cdot)$ and the boundedness of $\hat{A}$ and $W^{(L)}$, we have:

$$\|\nabla_{\tilde{Z}_i^{(L)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(L)}} \mathcal{L}\|_\infty \leq \rho \|\tilde{Z}_i^{(L)} - Z_i^{(L)}\|_\infty \leq \rho C^2 B^2 \varepsilon_H$$

Thus, the base case holds with $K = \rho C^2 B^2 \varepsilon_H$.

Similarly we can get $\|\nabla_{W_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{W_i^{(\ell)}} \mathcal{L}\|_\infty \leq K$.

Inductive Step: Assume that for some layer ℓ, the approximation error of the gradients is bounded by $K^{(\ell)}$:

$$\|\nabla_{\tilde{Z}_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(\ell)}} \mathcal{L}\|_\infty \leq K^{(\ell)}$$

$$\|\nabla_{W_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{W_i^{(\ell)}} \mathcal{L}\|_\infty \leq K^{(\ell)}$$

We need to show that the same bound holds for layer $\ell - 1$:

$$\|\nabla_{\tilde{Z}_i^{(\ell-1)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(\ell-1)}} \mathcal{L}\|_\infty \leq K^{(\ell-1)}$$

$$\|\nabla_{W_i^{(\ell-1)}} \tilde{\mathcal{L}} - \nabla_{W_i^{(\ell-1)}} \mathcal{L}\|_\infty \leq K^{(\ell-1)}$$

Using the chain rule and the Lipschitz continuity of $\sigma'(\cdot)$, we have:

$$\begin{aligned}
&\|\nabla_{\tilde{Z}_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(\ell)}} \mathcal{L}\|_\infty = \\
&\|\delta_i^{(\ell)} \hat{A}_i^{(\ell+1)} W_i^{(\ell)} \odot \sigma'(\tilde{Z}_i^{(\ell)}) - \delta_i^{(\ell)} \hat{A}_i^{(\ell+1)} W_i^{(\ell)} \odot \sigma'(Z_i^{(\ell)})\|_\infty \\
&\leq \|\hat{A}_i^{(\ell+1)} W_i^{(\ell)}\|_\infty \left(\|\delta_i^{(\ell)}\|_\infty + \|\sigma'(\tilde{Z}_i^{(\ell)}) - \sigma'(Z_i^{(\ell)})\|_\infty \right) \\
&\leq C^2 B^3 \left(K^{(\ell)} + \rho \|\tilde{Z}_i^{(\ell)} - Z_i^{(\ell)}\|_\infty \right) \\
&\leq C^2 B^3 \left(K^{(\ell)} + \rho C B \varepsilon_H \right)
\end{aligned}$$

So we can get $\|\nabla_{\tilde{Z}_i^{(\ell)}} \tilde{\mathcal{L}} - \nabla_{\tilde{Z}_i^{(\ell)}} \mathcal{L}\|_\infty \leq K^{(\ell-1)}$ by letting $K^{(\ell-1)} = C^2 B^3 \left(K^{(\ell)} + \rho C B \varepsilon_H \right)$

Similarly, we can get the inequality for the weight gradient.

Thus, by induction, the approximation error of the gradients is bounded by some K for all layers ℓ. □

1.3 Theorem: Convergence Guarantee

Theorem 5. *With the GNN model of L layers, given the local minimizer W^*, the initial weights $W_{(1)}$, and the staleness bound ε_H, suppose that: 1. The activation function $\sigma(\cdot)$ and the gradient $\nabla \mathcal{L}$ are ρ-Lipschitz continuous; 2. For the matrices $\hat{A}$, H, and W, all the gradients of the loss function with respect to the weight matrix $\|\nabla_W \tilde{\mathcal{L}}\|_\infty$, $\|\nabla_W \mathcal{L}\|_\infty$ and $\|\nabla \mathcal{L}(W)\|_\infty$ are bounded by some constant $G > 0$; 3. The loss $L(W)$ is ρ-smooth.*

Then there exists a constant $K > 0$ such that $\forall N > L\varepsilon_H$, if the distributed GNN is trained in parallel correspondingly with bounded staleness for $R \leq N$ iterations, where $R \in [1, \ldots, N]$ is chosen uniformly and the learning rate $\eta = \min\left\{\frac{1}{\rho}, \frac{1}{\sqrt{N}}\right\}$, we have:

$$\mathbb{E}\|\nabla \mathcal{L}(W(R))^2\|_F \leq \frac{2}{\sqrt{N}} (\mathcal{L}(W_{(R)}) - \mathcal{L}(W^*) + \frac{\rho K}{\sqrt{N}})$$

Proof. Let $\Delta_{(i)} = \nabla_{W_{(i)}} \tilde{\mathcal{L}} - \nabla \mathcal{L}(W_{(i)})$. By Lemma 4 and the ρ-smoothness of $\mathcal{L}(W)$, we have:

$$\begin{aligned}
\mathcal{L}(W_{(i+1)}) &\leq \mathcal{L}(W_{(i)}) + \langle \nabla \mathcal{L}(W_{(i)}), W_{(i+1)} - W_{(i)} \rangle \\
&\quad + \frac{\rho}{2} \eta^2 \|W_{(i+1)} - W_{(i)}\|_F^2 \\
&= \mathcal{L}(W_{(i)}) - \eta \langle \nabla \mathcal{L}(W_{(i)}), \Delta_{(i)} \rangle) - \eta \|\nabla \mathcal{L}(W_{(i)})\|_F^2 \\
&\quad + \frac{\rho}{2} \eta^2 (\|\Delta_{(i)}\|_F^2 + \|\nabla \mathcal{L}(W_{(i)})\|_F^2 + 2\langle \Delta_{(i)}, \nabla \mathcal{L}(W_{(i)}) \rangle) \\
&\leq \mathcal{L}(W_{(i)}) - (\eta - \frac{\rho}{2}\eta^2) \|\nabla \mathcal{L}(W_{(i)})\|_F^2 + \frac{\rho}{2}\eta^2 \|\Delta_{(i)}\|_F^2 \\
&\leq \mathcal{L}(W_{(i)}) - (\eta - \frac{\rho}{2}\eta^2) \|\nabla \mathcal{L}(W_{(i)})\|_F^2 + \frac{\rho}{2}\eta^2 K
\end{aligned}$$

Summing over N iterations and dividing by $N(\eta - \frac{\rho}{2}\eta^2))$, we obtain:

$$\mathbb{E}\|\nabla \mathcal{L}(W(R))^2\|_F^2 \leq \frac{2}{\sqrt{N}} (\mathcal{L}(W_{(1)}) - \mathcal{L}(W^*) + \frac{\rho K}{\sqrt{N}})$$

In particular, $\mathbb{E}\|\nabla \mathcal{L}(W(R))^2\|_F^2 \to 0$ when $N \to \infty$. The above concludes that the convergence is guaranteed.

In the context of sampling, the above analysis holds, the staleness bound ε_H ensures that the historical embeddings used in the sampling process are not too stale, thus maintaining the accuracy of the gradients and the convergence of the model. □